Brady Shaw

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Education

Utah State University Logan, UT

B.S. in Statistics, Minors in Mathematics and Psychology *Expected Dec. 2026*

- Relevant Coursework: Data Technologies (SQL, Web Scraping, XML), Modern Regression Methods, Applied Machine Learning & Multivariate Statistics, Analysis of Designed Experiments, Mathematical Statistics, Probability, Linear Algebra, Introduction to R. *In progress:* Python for Data Science, Data Structures & Algorithms (Java), Statistical Visualization, Mathematical Methods for Data Science

Projects

Olympic Medal & IMDb Analysis | *SQL, Web Scraping, Data Visualization* Fall 2025

- Engineered complex SQL queries with joins and subqueries to analyze Olympic medal data across 162 countries from multi-table relational databases
- Developed automated web scraping pipelines to extract and compare IMDb Top-250 movie ratings (2004 vs. 2025), discovering 61% showed improved ratings
- Applied regular expressions to clean and standardize 250+ movie titles from unstructured HTML data
- Created publication-ready visualizations and statistical reports to communicate data-driven insights

World Population Growth Estimation | *API Integration, Statistical Simulation* Fall 2025

- Built R functions to scrape real-time population data from web APIs and estimate annual growth rates with robust error handling
- Conducted Monte Carlo simulations with 10,000+ iterations to calculate confidence intervals, achieving estimates within 5% of official projections
- Automated data extraction from 64 XML files, parsed ISO 8601 timestamps, and performed time series analysis

Athlete Injury Risk Prediction | *Machine Learning, R, Statistical Modeling* Apr. 2026

- Built and cross-validated five supervised classification models (KNN, LDA, QDA, Naïve Bayes, Logistic Regression) in R to predict injury risk for 800 collegiate athletes from 18 physiological and performance features
- Tuned hyperparameters via 10-fold cross-validation and grid search, benchmarking results against teammates' SVM, decision tree, random forest, and neural network models
- Selected QDA as the final model based on validation performance (97.5% accuracy), achieving 96.25% accuracy on a held-out test set
- Conducted exploratory data analysis, feature standardization, and coefficient-level interpretation to surface actionable injury-risk insights

Experience

Kitchen Design Specialist Feb. 2024 – Present

Home Depot *Logan, UT*

- Promoted twice since joining in Feb. 2024 (Millworks Associate → Flooring Specialist → Kitchen Design Specialist)
- Design customer kitchens in 20/20 design software and generate detailed quotes, contributing to $150K+ in annual revenue
- Analyze customer requirements and translate technical specifications into actionable recommendations

Full-Time Missionary Nov. 2021 – Nov. 2023

The Church of Jesus Christ of Latter-day Saints *Wisconsin*

- Coordinated logistics and travel arrangements for 150+ missionaries using data-driven scheduling
- Led training programs and mentored new team members, strengthening leadership and communication skills

Technical Skills

Languages & Tools: R, SQL, Python (NumPy, pandas), XML, Web Scraping, Statistical Programming
Data Analysis: Regression Modeling, Forecasting, Monte Carlo Simulation, Hypothesis Testing, Exploratory Data Analysis, Machine Learning
Techniques: Data Cleaning, API Integration, Time Series Analysis, Regular Expressions, Data Visualization, Cross-Validation